

What Drives the International Transfer of Climate Change Mitigation Technologies? Empirical Evidence from Patent Data

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Abstract Technology transfer plays a key role in global efforts to reduce greenhouse gas emissions. In this paper, we characterize the factors that promote or hinder the international diffusion of climate-friendly technologies using detailed patent data from 96 countries for the period 1995–2007. The data provide strong evidence that lax Intellectual Property regimes have a strong and negative impact on the international diffusion of patented knowledge. Restrictions on international trade and foreign direct investment also hinder the diffusion of climate-friendly technologies. Surprisingly, local technological capabilities tend to discourage transfers. While broad indicators of technology capabilities are expected to facilitate transfers, this latter result stems from our technology-specific definition of local capabilities, which makes it possible to capture a substitution effect between local and foreign inventions.

Keywords Climate change · Technology diffusion · Technology transfer

JEL Classification O33 · O34 · Q54

1 Introduction

The international diffusion of technologies for mitigating climate change is at the core of current discussions surrounding the post-Kyoto regime. Technology development and diffusion are considered strategic objectives in the 2007 Bali Road Map. North-to-South technology transfer is of particular interest since technologies that have been developed mostly in

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industrialized countries are urgently required to mitigate greenhouse gas (GHG) emissions in fast-growing emerging economies. A recent study looking at patents filed in 13 climate-change mitigation technologies shows that two-thirds of the inventions patented worldwide between 2000 and 2005 have been developed in only three countries: Japan, the United States, and Germany (Dechezleprêtre et al. 2011).

Fostering technology transfer involves considerable policy and economic challenges because developing countries are reluctant to bear the financial costs of catching up on their own, while firms in industrialized countries refuse to give away strategic intellectual assets. This has led to an intense debate on policies that affect technology diffusion, with a particular focus on the role of intellectual property rights (IPRs), which developing countries view as a barrier to technology diffusion (ICSTD 2008). By contrast, industrialized countries argue that IPRs provide innovators with incentives to disseminate their inventions through market channels, such as foreign direct investment (FDI) and the international trade of equipment goods (Barton 2007). Industrialized countries argue that developing countries can promote technology transfers by increasing their capacity to absorb new technologies.

This paper examines these issues by identifying the factors that promote or hinder the international diffusion of climate-friendly technologies. We focus the analysis on the most relevant questions in current policy discussions. First, how important to technology transfer is the recipient countries' capacity to absorb foreign technologies? What is the impact of the level of enforcement of IP law on technology transfer? Do barriers to trade and FDI significantly reduce the transfer of technologies? What is the impact of domestic climate policies in the recipient countries on technology transfer? We also investigate whether the answers to these questions are specific to climate-friendly technologies.

We address these questions using a dataset of climate-related patents filed in 96 countries from 1995 to 2007, obtained from the World Patent Statistical Database (PATSTAT). We focus the analysis on ten climate-friendly technologies: three renewable energy technologies (wind, solar, and hydropower), three technologies related to energy conservation in buildings (energy-efficient lighting, thermal insulation, and energy-efficient heating), two emission reduction technologies in regular fossil fuel power generation (carbon capture and storage and "clean coal"), storage technology (fuel cells), and electric and hybrid vehicles. Although not all climate-friendly technologies are covered, our coverage spans various sectors, including transportation, electricity and heat production, manufacturing, and the residential sector. Moreover, we build a benchmark dataset that includes all patents filed in any technology to compare climate-friendly technologies with other patented technologies.

As a measure of diffusion, we count the number of patent applications in recipient countries for technologies invented abroad. This indicator is a proxy of cross-border technology transfer because holding a patent in a country gives the holder the exclusive right in that country to commercially exploit the technology. The literature (see e.g., Keller 2004) identifies two main channels driving such patent transfers,¹ namely the trade of manufactured products (mostly machines and equipment which embody technology), and foreign direct investment (FDI) whereby multinational enterprises transfer firm-specific technology to their foreign affiliates.

Prior studies indicate that technology transfers through either channel involve patent filings in the recipient country, and therefore positively depend on the quality of its patent system (Maskus 2000; Smith 2001; Ivus 2010). Smith (2001) moreover highlights a possible substitution effect between both channels depending on the strength of patent protection. To

¹ The licensing of patents is a third possible channel. Yet transfers via the latter is of much smaller magnitude in practice compared to trade and foreign direct investment, particularly for the environment-related technologies in which we are interested (see discussion *infra* in Sect. 4).

account for these mechanisms, we develop as a first step a theoretical model which we then test empirically to see how policy variables such as trade barriers, capital control and the strength of patent protection influence aggregate flows of environment-related patents.

Our empirical results confirm that weak patent protection has a strong and negative influence on the international diffusion of patented knowledge for our ten technology classes of low-carbon technologies. The effect of restrictions to international trade and foreign direct investment is negative as well, but with a smaller order of magnitude. We also highlight a positive impact of climate policies adopted in recipient countries and, more surprisingly, a negative effect of local technological capabilities on the import of technology. This latter finding denotes the fact that local capabilities in restricted technology classes actually reflect domestic innovation, which can be substituted for imported technology. Finally, results from our benchmark dataset indicate that climate technologies do not respond differently than the average technology to changes in trade, IP or FDI policies. This suggests that there is little ground to design policies specifically targeted at climate-related technologies in this area, at least on the supply side.

The literature dealing with the international diffusion of environment-related technology is limited but is growing rapidly.² Unlike the present work, this literature is mostly descriptive. [Lanjouw and Mody \(1996\)](#) presented the first patent-based empirical evidence for the international diffusion of environmentally responsive technology. Focusing on chlorine-free technology in the pulp and paper industry, [Popp et al. \(2007\)](#) provide evidence that environmental regulation may promote international technology transfer. They observe for instance an increase in the number of patents filed by US inventors in Finland and Sweden after passage of tighter regulations in these countries. Several case studies discuss whether stricter patent protection promotes or hinders the transfer of climate-related technology to developing countries (see, for example, [Barton 2007](#); [Ockwell et al. 2008](#)). Finally, PATSTAT data was recently used to describe the geography of innovation and international technology diffusion ([Dechezleprêtre et al. 2011](#)).

This study is one of the first econometric studies dealing with the transfer of environmental technologies. There are only a few previous studies in this field, such as the paper by [Dekker et al. \(2012\)](#) that assesses how sulphur protocols trigger invention and diffusion of technologies for reducing SO₂ emissions. A paper by [Hascic and Johnstone \(2011\)](#) is also related to our work. They use a similar dataset to study the impact of the Kyoto protocol on technology diffusion. Our focus differs from previous studies because we deal with a broader set of policy variables, including trade barriers and FDI control. Another study by [Verdolini and Galeotti \(2011\)](#) investigates the drivers of innovation and diffusion in energy-efficient technologies.

As a measure of diffusion, our approach based on patent counts is similar to that of [Lanjouw and Mody \(1996\)](#), [Eaton and Kortum \(1999\)](#), or [Hascic and Johnstone \(2011\)](#). Other studies have used patent citations to measure the extent of international knowledge flows (see [Jaffe et al. 1993](#); [Peri 2005](#)), with one important difference. Indeed, inventors file patents abroad to reap private benefits. Therefore, while citations made by inventors to previous patents are an indicator of *knowledge spillovers*, our indicator is a proxy for *market-driven knowledge flows*.

The paper is organized as follows: Sect. 2 discusses the use of patents as indicators of technology transfer. The dataset is presented in Sect. 3. In Sect. 4, we develop a theoretical model that describes the diffusion of inventions between countries to derive predictions

² By contrast, the general empirical literature on international technology diffusion is well developed (for a good survey, see [Keller 2004](#)).

regarding the impact of trade barriers, FDI control, and the stringency or laxity of IPRs on technology transfer. Econometric models and results are described in Sect. 5. A final section summarizes the main results.

2 Patents as indicators of technology transfer

In the empirical literature, scholars propose a number of solutions for the measurement of international technology transfers. Because international trade and FDI are major channels for the transmission of technology knowledge across countries, many studies use the inflows of intermediate goods or FDI as proxy variables for international technology transfer (for example, [Coe and Helpman 1995](#); [Lichtenberg and van Pottelsberghe de la Potterie 2000](#)). Data on trade and FDI are readily available for a large number of countries, thereby allowing for a broad geographical coverage. However, these data are highly aggregated in terms of economic sectors, which prevents their use for measuring the flows of climate-friendly technologies. In addition, trade and FDI are only indirect vehicles of knowledge transfer.

The use of patent data has gained popularity in the recent empirical literature,³ Patent data focus on outputs of the inventive process ([Griliches 1990](#)). Furthermore, they provide a wealth of information on the nature of the invention and the applicant. Most importantly, they can be disaggregated to specific technological areas. Finally, they indicate not only the countries where inventions are being made, but also where these new technologies are being used. These properties make our study of climate-change mitigation technologies possible. Of course, patent data also present drawbacks, which we discuss below.

Patent data is an effective vehicle for analyzing where inventions are being made and used, which can be demonstrated by considering a simplified version of the innovative process.. In the first stage, an inventor from country i develops a new technology. The inventor then decides to patent the new technology in certain countries. A patent in country j grants the inventor the exclusive right to commercially exploit the innovation in that country. Patenting in country j indicates the inventor plans to use it there. The set of patents protecting the same invention in several countries is called a patent family.

In this paper, we use the number of patented inventions filed in country j by inventors from country i as an indicator of the number of innovations transferred from country i to country j . As mentioned in the introduction, this indicator has already been used in previous studies (see, for instance, [Lanjouw and Mody 1996](#); [Eaton and Kortum 1999](#)). This indicator differs, however, from indicators based on backward patent citation used in the literature measuring knowledge spillovers (see [Jaffe et al. 1993](#)).⁴

Our indicator is obviously not without drawbacks. The first limitation is that for protecting innovations, patents are only one of several means, along with lead-time, industrial secrecy, or purposefully complex specifications ([Cohen et al. 2000](#); [Frietsch and Schmoch 2006](#)). In fact, inventors may prefer secrecy to avoid the public disclosure of the invention (which patent law requires) or to save the significant fees attached to patent filing. Despite such

³ Alternatively, [Branstetter et al. \(2006\)](#) and [Smith \(2001\)](#) use royalty payments and licenses. Such data provide an accurate view of the commercial value of technology transfers through a particular channel, namely IP licensing, but those data are available only for the United States. Therefore, IP licensing is not appropriate for assessing global technology transfers through various channels.

⁴ It is argued that the number of forward citations reflects the value of individual patents. This has been exploited in the literature to compute weighting coefficients. We could have done the same to control for the heterogeneity of patents' value. However, citation data are not available for most countries (with the exceptions of the United States and the European Union).

reservations, there are few examples of economically significant inventions that have not been patented ([Dernis and Guellec 2001](#)), although the propensity to patent differs between sectors, depending on the nature of the technology ([Cohen et al. 2000](#)) and the risk of imitation in a country. The factors that influence the propensity to patent have a significant effect on our data because patenting is more likely in countries that have strong technological capabilities and that strictly enforce IPRs. The econometric models presented below partly control for these limitations.

A further drawback of our indicator is that certain forms of knowledge are not patentable. Know-how or learning-by-doing, for example, cannot be easily codified because these are skills embodied in individuals (tacit knowledge). Nevertheless, research has shown that flows of patented knowledge and of tacit knowledge are positively correlated ([Cohen et al. 2000](#); [Arora et al. 2008](#)).

Another limitation is that a patent grants the exclusive right to use the technology only in a given country; however, this does not mean that the patent owner will actually use the technology in that country. This limitation could significantly bias our results. For example, if applying for patent protection did not cost anything, then inventors might patent widely and indiscriminately. However, in practice this is not the case. [Dechezleprêtre et al. \(2011\)](#) show that the average invention is patented in only two countries.⁵ Patenting is costly, in both the preparation of the application and the administration associated with the approval procedure (see [Helfgott 1993](#)). In addition, possessing a patent in a country is not always in the inventor's interest if that country's enforcement is weak. Additionally, publication of the patent in a local language can increase vulnerability to imitation (see [Eaton and Kortum 1996, 1999](#)). Therefore, inventors are unlikely to apply for patent protection in a country unless they are relatively certain of the potential market for the technology covered. Finally, because patenting protects an invention only in the country where the patent is filed, inventors are less likely to engage in strategic behavior to protect their inventions and prevent the use of their technology in the production of goods imported by foreign competitors into their domestic markets.

Another problem is that the economic value of individual patents is heterogeneous and its distribution is skewed. Therefore, because many patents have little value, the number of patents does not accurately reflect the value of innovations. This accuracy problem is less of a concern for this paper because we focus on international diffusion or exported technologies, which are of the highest value and make up only about a quarter of all inventions ([Lanjouw et al. 1998](#)). A possible solution to this problem would be to weight patents by their forward citations, but citation data is not yet available for all countries.

3 Patent data

Over the past several years, the EPO, and the OECD's Directorate for Science, Technology and Industry have developed a worldwide patent database—the EPO/OECD PATSTAT. PATSTAT is unique in that it covers more than 80 patent offices and contains around 70 million patent documents. PATSTAT data have not been exploited extensively until recently because they only recently became available. Our study is the first to use PATSTAT data to explain the diffusion of climate-change mitigation technologies.

We extracted all the patents filed worldwide in 10 climate-change mitigation fields, the precise description of which can be found in Table 1. Our patent data dates back as far as 1861

⁵ In fact, about 75 % of inventions are patented in only one country.

Table 1 Description of the technology fields covered

Technology field	Description of aspects covered
CCS	Extraction, transportation, storage, and sequestration of CO ₂
Insulation	Elements or materials used for heat insulation; double-glazed windows; energy recovery systems in air conditioning or ventilation
Electric and hybrid vehicles	Electric propulsion of vehicles; regenerative braking; batteries; control systems specially adapted for hybrid vehicles
Clean coal	Efficiency improving fossil fuel technologies for electricity generation: coal gasification, improved burners, fluidized bed combustion, improved boilers for steam generation, improved steam engines, super-heaters, improved gas turbines, combined cycles, cogeneration
Fuel cells	Fuel cells (electrochemical generators wherein the reactants are supplied from outside); manufacture of fuel cells
Hydro	Hydro power stations; hydraulic turbines; submerged units incorporating electric generators; devices for controlling hydraulic turbines
Lighting	Compact fluorescent lamps; electroluminescent light sources (LED)
Solar	Solar photovoltaic (conversion of light radiation into electrical energy), including solar panels; concentrating solar power (solar heat collectors having lenses or reflectors as concentrating elements); solar heat (use of solar heat for heating & cooling)
Heating	Heat pumps, central heating systems using heat pumps; energy recovery systems in air conditioning
Wind	Wind motors; devices aimed at controlling such motors

for some countries.⁶ This data represents 826,672 patent applications filed in 96 countries.⁷ On average, climate-related patents included in our dataset represent less than 1 % of the total annual number of patents filed worldwide.

Patent applications related to climate-change mitigation are identified using the International Patent Classification (IPC) codes and the European classification codes (ECLA) available in PATSTAT. To identify the relevant IPC classes, we rely on the previous work of the OECD and the EPO. The list of IPC and ECLA codes for climate-change mitigation technologies is now readily available online.⁸ In addition to climate-friendly patents, we also use other data, in particular, the data used to describe policies that may influence international patent transfers. These data are described in Sect. 5.

4 Modeling transfer channels

The ultimate goal of our study is to explain the technology flows between a pair of countries. In practice, these flows occur through international trade of manufactured products or through FDI. Licenses to unaffiliated firms represented less than 0.1 % of the total value of licenses, FDI, and exports of manufactured products from the United States to the rest of the world in

⁶ Note that our estimations only span the years 1995–2007. However, we can use data as far back as 1861 to construct the country-specific patent stocks.

⁷ Note that Least Developed Countries are not present in our dataset, for two related reasons: their patenting activity is extremely limited and available statistics are not reliable.

⁸ A list of environment-related patent classification codes is available from the OECD's Environmental Policy and Technological Innovation (EPTI) website: www.oecd.org/environment/innovation. We gratefully acknowledge the continuous efforts of Nick Johnstone and Ivan Hascic to provide updated classification codes to the research community.

1989 (Smith 2001). Moreover, Anand and Khanna (2000) find that about 68 % of licensing contracts take place in only two sectors—chemicals and drugs (46 %) and electronics and electrical equipment (22 %)—of which neither overlaps with the climate-mitigation technologies we examine in this paper. A recent study on the Chinese solar photovoltaic industry also confirms that patent licensing does not play any role in this sector, the key vectors are FDI and the trade of manufacturing equipment (De la Tour et al. 2011). We can thus focus the entire analysis on trade and FDI.

Among the possible drivers of international flows, we are interested in testing the effects of such policy variables as trade barriers, capital controls, or the strength of patent law in the recipient country that affect these channels. A problem is that the relationship between these variables is complex because FDI and trade are partly substitutes in technology transfer.

There is no doubt that the use of the FDI and trade transfer channels positively depends on the strength of IPRs in the recipient country, but not with equal intensity (Maskus 2000; Smith 2001; Ivus 2010). This is due to the fact that each channel does not bring about the same level of technology transfer, nor does it yield the same risk of technology imitation. As stated by Smith (2001), exports of manufactured products bring about lower flows of technology than does FDI, because technology transfers concern only product innovations, while process innovation remains in the originating country. By contrast, FDI implies that the manufacturing technology is also transferred to the recipient country. Therefore, the risk of being imitated is higher under FDI because local competitors can access the technology through reverse engineering of products as well as through the circulation of labor in the recipient country. The FDI channel is thus more affected by the strictness of IPRs.

These differences between the FDI and trade transfer channels have two consequences. First, as shown empirically by Smith (2001), companies tend to substitute FDI for exports when patent protection increases in the recipient country. Second, a FDI is likely to generate more patent applications than exports in the recipient country because corporations transfer not only product innovations but also manufacturing technology.

To characterize these complex mechanisms thoroughly, we develop a model that we use to derive predictions that we will test in the next section. The model features a set of firms that decide whether to transfer a technology into a foreign country and what transfer channel (trade or FDI) to use.

There are K_i heterogeneous firms in country i . Each has developed an innovative technology they seek to commercialize in country j . The technology can serve a market of size θ_j in country j , which is drawn from a distribution F_j on the interval $[0, \theta_j^{max}]$. To simplify notations, we ignore the subscripts i and j in the following.

Innovative technology is defined as an information set $\{a, b\}$ where a and b denote, respectively, a product invention and the related production process; both of which are patentable. If the firm with the technology of type θ decides to enter country j , it can choose between two channels: (1) manufacturing the innovative product at home and exporting it into country j or (2) investing directly in country j to set up a local production unit. In the first case, the manufacturing process b remains in country i , but competitors in country j can access the product invention a through reverse engineering. In the second case, both the product and manufacturing process are transferred into country j . In addition to reverse engineering, local competitors also can access the manufacturing process (e.g., through labor circulation in the local labor market).

Assuming that a firm with type θ decides to enter country j , its expected profit is

$$V_\gamma(\theta) = p_\gamma(ipr)\theta\pi_\gamma - c_\gamma \text{ with } \gamma = I, T \quad (1)$$

where I and T denote, respectively, the FDI and trade channels for the transfer of technology. The other parameters are the profitability of the foreign market π_γ , a fixed cost of entry c_γ , and $p_\gamma(ipr)$, which is the probability that the technology will not be counterfeited by local competitors. This probability depends on ipr , the stringency of patent law in a country, with $ipr \in [0, \infty]$ and $p_\gamma(ipr) \rightarrow 1$ when $ipr \rightarrow +\infty$.

We now introduce assumptions that aim to capture two key differences between the FDI and trade channels for the transfer of technology. First, we assume that exports entail a lower risk of imitation than FDI and are less responsive to patent strength, following previous papers (Smith 2001). Our first assumption is as follows:

Assumption 1 For any ipr , $p_T(ipr) > p_I(ipr) > 0$; $0 < p'_T(ipr) < p'_I(ipr)$.

The second difference between the FDI and trade channels concerns the costs and benefits of each channel. We assume a higher entry cost if technology transfer takes place through FDI as it requires investing upfront in a new production unit. However, FDI also make it possible to reduce the variable cost of production as exporting goods entails the additional risk of variability in transportation costs, exchange rates, trade tariffs and, in some cases, higher manufacturing costs. Accordingly, we introduce the following assumption:

Assumption 2 The costs and profitability of trade and FDI are such that $c_I > c_T$, $\pi_I > \pi_T$, $\pi_I/c_I > \pi_T/c_T$.

Under these two assumptions, the choice between the FDI and trade channels depends on a tradeoff between the lower cost of entry through trade and the economies of scale than can be achieved through FDI if the market is sufficiently large (if θ is high). The last inequality imposes that the trade channel is preferable to the FDI channel if the market is not sufficiently large.

From Assumptions 1 and 2 and Eq. (1) directly follows the entry strategy of the firm:

Lemma 1 *The firm does not transfer the technology if $\theta < \theta_0$. It transfers the technology through trade if $\theta_0 \leq \theta < \theta_1$ and through FDI if $\theta \geq \theta_1$ with the following:*

$$\theta_0 = \frac{c_T}{p_T(ipr)\pi_T} \quad \text{and} \quad \theta_1 = \frac{c_I - c_T}{p_I(ipr)\pi_I - p_T(ipr)\pi_T}$$

Based on Lemma 1, we will now derive N_{ij} , the number of patent flows from country i to country j . Let α and β denote the number of patents filed by a firm when using the trade or the FDI channel, respectively. Because FDI requires transferring both product and process inventions, while trade only requires transferring the former, we have $\alpha < \beta$. Assuming without loss of generality that ipr is large enough to have some FDI ($\theta^{\max} \geq \theta_1$), it follows that the number of firms choosing each channel is:

- FDI: $K[1 - F(\theta_1)]$
- Trade: $K[F(\theta_1) - F(\theta_0)]$
- No entry: $K F(\theta_0)$

The number of patents filed in country j by inventors from country i is therefore as follows:

$$N_{ij} = K(\alpha F(\theta_1) - F(\theta_0)) + (\beta(1 - F(\theta_1)))$$

or, after rearranging, as follows:

$$N_{ij} = K(\beta - F(\theta_1)(\beta - \alpha) - \alpha F(\theta_0))$$

where $F(\theta_1)(\beta - \alpha)$ captures a substitution effect between export and FDI, while $\alpha F(\theta_0)$ captures a barrier to entry effect. This expression makes it possible to derive general predictions about the expected effects of policy variables, such as the strength of patent law in country j (ipr) or barriers to trade or FDI in country j (reflected in c_I and c_T), as follows:

Proposition 1 *The policy variables have the following impact on the aggregate flows of patents from country i to country j :*

1. *Stronger patent protection in country j increases the incoming flow of patents*
2. *Higher barriers to FDI in country j decrease the incoming flow of patents.*
3. *Higher barriers to trade in country j have an ambiguous effect on the incoming flow of patents.*

Proof We start by looking at the impact of ipr , c_I and c_T on the threshold values of θ . We then arrive at the following:

$$\begin{aligned} \frac{\partial \theta_0}{\partial ipr} &= -\frac{c_T}{(p_T)^2 \pi_T} p'_T < 0; \quad \frac{\partial \theta_0}{\partial c_T} = \frac{1}{p_T \pi_T} > 0; \quad \frac{\partial \theta_0}{\partial c_I} = 0; \quad \frac{\partial \theta_1}{\partial ipr} \\ &= -\frac{(c_I - c_T)(p_I' \pi_I - p_T' \pi_T)}{(p_I \pi_I - p_T \pi_T)^2} < 0; \quad \frac{\partial \theta_1}{d\theta} = -\frac{1}{p_I \pi_I - p_T \pi_T} < 0; \quad \frac{\partial \theta_0}{\partial c_I} = \frac{1}{p_I \pi_I - p_T \pi_T} > 0. \end{aligned}$$

We can then differentiate N_{ij} with respect to each policy variable in order to assess their impact on the number of patent flows from country i to j . We derive $\frac{\partial N_{ij}}{\partial ipr} = -K \alpha f(\theta_0) \frac{\partial \theta_0}{\partial ipr} + K(\alpha - \beta) f(\theta_1) \frac{\partial \theta_1}{\partial ipr}$ is positive as $\alpha - \beta < 0$. The sign of $\frac{\partial N_{ij}}{\partial c_T} = -K \alpha f(\theta_0) \frac{\partial \theta_0}{\partial c_T} + K(\alpha - \beta) f(\theta_1) \frac{\partial \theta_1}{\partial c_T}$ is ambiguous as the first term is negative, while the second is positive. Finally, $\frac{\partial N_{ij}}{\partial c_I} = K(\alpha - \beta) f(\theta_1) \frac{\partial \theta_1}{\partial c_I} > 0$.

As expected, the model predicts that strong patent protection fosters incoming patent flows as it promotes entry through exports and substitutes patent-intensive FDI for less patent-intensive exports. This substitution effect similarly explains why barriers to FDI negatively impact patent flows. That is, firms in country i react to higher FDI barriers by substituting trade for FDI. By contrast, with regard to the impact of trade barriers on patent flows, the net effect of substitution is ambiguous. Trade barriers obviously hinder technology transfer through exports, but then transfers through FDI could serve as substitutes for exports. As FDI is more patent intensive, this may lead to increased patent flows across countries. $\square$

5 Empirical issues

We have constructed a panel dataset for each of the 10 technology fields described in Sect. 3. This data allow us estimating the model on each field allows us to control for technology-specific factors and to test the robustness of our results across a wide range of technologies.

5.1 Estimation equations

Our dependent variable, N_{ijt} , is the number of patents filed in country j by inventors from country i . As it only has non-negative integer values, we will assume that N_{ijt} follows a Poisson distribution. Accordingly, the single parameter of the distribution λ_{ijt} is the expected value of N_{ijt} , given a set of explanatory variables. More specifically, we adopt the following specification:

$$N_{ijt} = \exp(\alpha \ln k_{jt-1} + \beta_1 \text{ipr}_{jt-1} + \beta_2 \text{tariff}_{jt-1} + \beta_3 \text{fdi_control}_{jt-1} + \beta_4 \text{policy}_{jt-1} \\ + \gamma I_{it-1} + \delta \ln GDP_{jt-1} + \psi_{ij} + \chi' \varphi_t + u_{ijt})$$

Here, k_{jt-1} describes the recipient country's stock of knowledge, ipr_{jt} , tariff_{jt} fdi_control_{jt} which are variables describing the strictness of patent law, the size of the tariffs restricting imports, and the barriers to FDI in the recipient country in year t , respectively. The variable policy_{jt} captures the strictness of climate policies in the recipient country; I_{it-1} is the number of relevant inventions from the source country available for potential transfer; GDP_{jt} is the country j 's growth domestic product; ψ_{ij} is a time-invariant fixed effect; φ_t is a set of time dummies; and u_{ijt} is a random term capturing unobserved heterogeneity, such as the differences in countries' propensity to patent. The parameters to be estimated are α , β_1 , β_2 , β_3 , β_4 , γ , δ and χ . We now describe in detail how we constructed the main independent variables.

The recipient country's stock of knowledge (k_{jt-1})

We seek to understand whether technology capabilities in the recipient country have an influence on the transfers of patents. This leads us to use the variable k_{jt-1} , which is the discounted stock of previously filed patents in the same technology area at date $t - 1$ by local inventors in the recipient country j . Note that while local absorptive capabilities are usually measured with broad cross-technology indicators (e.g., education or number of patented inventions in a given country), this variable provides a very specific indicator of local technology capabilities for each technology. By contrast with more generic variables, it has thus an ambiguous effect because it also captures a potential competition effect between imported patents and patents already available in the recipient country.

Following [Peri \(2005\)](#), the patent stock is calculated using the perpetual inventory method. We initiate the stock for the year 1950⁹ and use the recursive formula

$$k_{jt-1} = (1 - d)k_{jt-2} + N_{j,t-1}$$

where $N_{j,t}$ is the number of patented technologies invented by domestic inventors in year t . The value chosen for d , the depreciation of R&D capital, is 15 %, a value commonly used in most of the literature (see [Keller 2002](#)); however, we check the robustness of our results using lower or higher discount rates.

The strictness of patent law (ipr_{jt})

The variable ipr_{jt} is a country-specific index built by [Park and Lippoldt \(2008\)](#). The value lies between 0 and 10¹⁰ and measures the strictness of intellectual property rights in the recipient country. As argued above, lax patent systems can deter the international trade of technologies because of the fear of counterfeiting (see, for example, [Maskus 2000](#); [Smith 2001](#); [Barton 2007](#)). The IP issue is hotly debated in the political arena.

The variable ipr_{jt} relates to the propensity to patent in country j , which may make our results more difficult to interpret. Stricter IPRs can lead to additional patent applications for technologies that would probably have been transferred through trade or FDI. Using ipr_{jt} can thus lead us to an overestimation of the effect of IPRs on technology transfer.

Barriers to trade (tariff_{jt})

The variable tariff_{jt} captures the existence of potential barriers to international trade. We use the "Taxes on international trade" 0–10 index from the Economic freedom of the world 2010

⁹ We arbitrarily set the value of the patent stock in 1950 at 0. Note that the influence of the initial stocks is infinitesimal as we start estimations in 1995.

¹⁰ The original index has values between 0 and 5, but we rescale it to ensure consistency with the trade and FDI variables, which take values between 0 and 10.

report. This index has three sub-components: revenues from trade taxes as a share of exports and imports as provided by the International Monetary Fund, the mean tariff rate as reported by the World Trade Organization, and the standard deviation of tariff rates (compared to a uniform tariff as wide variation in tariff rates exerts a more restrictive impact on trade). Proposition 1 tells us that restrictions to trade may have an ambiguous effect on the transfer of technologies embodied in capital equipment goods.

Barriers to FDI($fdi_control_{jt}$)

The variable $fdi_control_{jt}$ is an index of international capital market controls based on data from the International Monetary Fund (IMF).¹¹ The IMF reports on up to 13 different types of international capital controls. The zero-to-10 rating is the share of capital controls levied as a share of the total number of capital controls listed multiplied by 10.

The strictness of climate policies ($policy_{jt}$)

Domestic climate policies in the recipient country increase the demand for climate technologies. Given the scope of our country dataset, it is extremely difficult to find a variable that measures the strictness of each recipient's country climate policy in a comparable way. If available, country- and sector-specific carbon prices would be an ideal variable here. But in its absence, as a second-best option, we count the number of climate change related policies in place in each country using the International Energy Agency Climate Change Policies and Measures database.¹² While the explanatory power of this variable is admittedly limited, we expect that the implementation of a new climate-related policy should increase technology inflows. Counting the number of climate change related policies is comparable to the use of dummy variables to represent the introduction of new policies as is done, for instance, by Johnstone et al. (2010).

The number of source country inventions available for potential transfer $I_{i,t-1}$

$I_{i,t-1}$ is the number of inventions patented by inventors from country i anywhere in the world in year $t - 1$, and not previously patented. Any invention patented in several countries is therefore counted only once. This variable measures the number of inventions available for transfer from country i to country j in year i . *Ceteris paribus*, more patents should be transferred from countries that have a higher number of technologies available to be patented in foreign markets.

5.2 Estimation technique and sample

As explained above, we use a Poisson estimation because the number of patents transferred is a count variable. We could also use a negative binomial, which relaxes the assumption that the mean of the error term is equal to its variance. But, as we compute standard errors that are robust to heteroskedasticity and clustered by country pair in all our models,¹³ the exact functional form of the error distribution is not crucial. In order to deal with fixed effects at the country-pair level, we follow the approach of Hausman et al. (1984) and estimate by conditional maximum likelihood.¹⁴ Our panel runs from 1995 to 2007¹⁵ and includes 11,766

¹¹ The index of international capital market controls is available from the Economic Freedom of the World 2010 Annual Report. For both trade barriers and FDI controls, missing years were filled by interpolation.

¹² Available at <http://www.iea.org/textbase/pm/index.html>.

¹³ Note that STATA cannot produce robust and clustered standard errors for the fixed effect negative binomial estimator (although one can always bootstrap standard errors).

¹⁴ This is implemented by the `xtpoisson, fe` command in STATA.

¹⁵ 1995 is the first year for which we have data on trade policies, while 2007 is the last reliable year in the September 2009 version of the PATSTAT database.

Table 2 Descriptive statistics for independent variables

Variable	Observations	Mean	SD	Min	Max
P_{ij} ,	Depends on the technology (see Table 4)				
k_{jt-1}	Depends on the technology				
ipr_{jt}	131,175	6.95	1.77	2.16	9.76
$tariff_{jt}$	127,359	7.62	1.54	0.90	10.00
$fdi_control_{jt}$	127,359	4.94	3.28	0.00	10.00
$policy_{jt}$	62,964	8.68	16.46	0.00	147.00
I_{it-1}	Depends on the technology				
GDP_{jt}	125,610	5.17	1.60	1.87	9.42

Table 3 Descriptive statistics for the dependent variable, by technology

Technology	Observations	Mean	SD	Min	Max
CCS	11,177	2.14	5.72	0	83.80
Clean coal	20,005	5.26	17.39	0	254.50
Fuel cells	16,254	9.36	37.65	0	906.33
Electric/hybrid	13,728	6.17	25.74	0	526.28
Heat pumps	10,989	1.41	4.11	0	53.00
Hydro	10,451	0.70	1.92	0	28.00
Insulation	12,906	2.35	7.33	0	103.86
Lighting	15,131	9.84	37.11	0	1120.00
Solar	18,300	2.48	9.08	0	142.30
Wind	16,953	2.19	9.33	0	173.17

country pairs over that period. Note that because we use a fixed-effects Poisson estimator, the final estimation samples include fewer country pairs because the number of patent transfers between some country pairs is always equal to zero, and these pairs get dropped from the estimation sample. Tables 2 and 3 present summary statistics for the dependent and independent variables used in the estimations.

6 Results

We report the main results in Table 4. The quality of the estimations is very good: the pseudo R-square¹⁶ ranges between 0.79 and 0.94. Estimates across technologies are relatively similar: coefficients exhibit the same signs, although the size may differ. This homogeneity allows a common interpretation. Control variables exhibit expected signs: the number of inventions from the source country available for potential transfer and GDP_{jt} , which controls for the size of the recipient country, both raise technology flows. We now focus the discussion on six policy-relevant questions.

¹⁶ To calculate the R2 we re-estimate all models using unconditional pseudo maximum likelihood (i.e., we include a full set of country-pair dummies). To deal with convergence problems in the presence of many zeros and many dummies we use the `ppml` command in Stata (Santos Silva and Tenreiro 2011).

Table 4 Main results

Technology	(1) CCS	(2) Clean coal	(3) Fuel cells	(4) Electric & hybrid	(5) Heating	(6) Hydro	(7) Insulation	(8) Lighting	(9) Solar	(10) Wind
$\ln k_{jt-1}$	-0.301*** (0.071)	-0.228* (0.117)	-0.088 (0.068)	-0.246*** (0.071)	-0.163 (0.099)	-0.206*** (0.077)	-0.212*** (0.072)	-0.161 (0.103)	-0.223*** (0.068)	-0.161*** (0.062)
i_{prjt}	0.416*** (0.066)	0.435*** (0.070)	0.588*** (0.070)	0.606*** (0.094)	0.492*** (0.076)	0.269*** (0.075)	0.462*** (0.052)	0.514*** (0.081)	0.519*** (0.054)	0.447*** (0.081)
$fdi_control_{jt}$	-0.063*** (0.017)	-0.067*** (0.015)	-0.071*** (0.014)	-0.056*** (0.017)	-0.077*** (0.018)	-0.078*** (0.015)	-0.069*** (0.011)	-0.065*** (0.017)	-0.068*** (0.014)	-0.045*** (0.013)
$tari_{ffjt}$	-0.036 (0.041)	-0.124** (0.048)	-0.111** (0.048)	-0.029 (0.060)	-0.142** (0.064)	-0.106*** (0.039)	-0.077* (0.040)	-0.147*** (0.046)	-0.046 (0.040)	-0.046 (0.036)
$policy_{jt}$	-0.000 (0.002)	-0.001 (0.002)	0.008*** (0.002)	0.003** (0.001)	0.000 (0.002)	0.009*** (0.002)	0.001 (0.002)	0.003* (0.002)	-0.003 (0.002)	0.008* (0.004)
I_{t-1}	0.460*** (0.041)	0.504*** (0.059)	0.717*** (0.043)	0.428*** (0.057)	0.888*** (0.046)	0.875*** (0.041)	0.497*** (0.038)	0.703*** (0.066)	0.756*** (0.051)	1.097*** (0.042)
$\ln GDP_{jt}$	0.605 (0.600)	1.239** (0.489)	0.719 (0.443)	1.074** (0.518)	0.248 (0.507)	0.648 (0.441)	1.082*** (0.331)	1.070 (0.667)	0.306 (0.481)	0.146 (0.438)
Observations	9,295	1,634	1,364	1,171	9,405	8,877	1,0912	1,281	1,541	1,4542
Country pairs	845	1,494	1,244	1,071	855	807	992	1,171	1,401	1,322
R2	0.85	0.89	0.93	0.94	0.82	0.79	0.92	0.91	0.90	0.94

* Significant at the 10 % level, ** significant at the 5 % level, *** significant at the 1 % level. The dependent variable is the number of patents transferred from country i to country j in year t . All columns are estimated using a fixed-effects Poisson and include a full set of year dummies (not reported for brevity). Standard errors clustered at country-pair level in parentheses

6.1 Does accumulated knowledge facilitate the import of technology?

The variable k_{jt-1} has a significantly negative impact in 7 out of 12 regressions. Although this negative sign might seem at odds with the general idea that high technology absorptive capacities facilitate incoming technology transfers, it makes economic sense given the technology-specific indicator that we use. Indeed, previous inventions in a given technology area could either be complements or substitutes for imported inventions in that same area. They are complements if they improve the local capacity to absorb new inventions; they are substitutes if they compete with imported inventions. Table 4 shows that the competition effect prevails. However, the aggregate effect is moderate: elasticities range between -0.16 to -0.30 .¹⁷ That is, a 10 % increase of k_{jt-1} , which is large as it is a stock variable, leads to 1.6–3.0 % fewer patents.

6.2 Do strict IPRs promote technology transfer?

As mentioned in introduction, this issue is very high in the political agenda. Our results confirm Proposition 1's predictions, namely that a significant positive impact of strict IPRs on technology transfer in all regressions.

The impact of strict IPRs on technology transfer is large, that is, an increase by one unit of the zero-to-ten rating brings about between 27 and 60 % more patent imports.¹⁸ Note that we may over-estimate the influence of IPR on *actual* technology transfers because a portion of the patenting could reflect a substitution between patented and non-patented knowledge flows, rather than additional technology flows.

6.3 Do trade barriers hinder technology transfer?

Proposition 1's theoretical predictions were ambiguous with regard to the role of trade barriers in reducing the transfer of knowledge through trade. However, this effect could be compensated for by the substitution of trade by FDI, which is a more patent-intensive channel. Estimations show that trade outweighs the FDI, that is, higher tariff rates have a statistically significant negative impact on patent flows in six regressions. It suggests that transferred technologies are frequently incorporated in equipment goods.

It is possible to compare the relative impact of trade barriers versus IPR by looking directly at the size of the coefficients because the two variables have zero-to-ten ratings. Trade barriers have much less influence, that is, a one unit increase leads to 7–15 % fewer imports, whereas the positive impact of IPR strictness on patent inflows lies in the range of 27–60 %.

6.4 Do restrictions on FDI hinder technology transfer?

We confirm Proposition 1's predictions, namely that controls over FDI significantly hinder patent imports in all regressions by 4.5–8 % for a one-unit increase of the FDI variable. Again, the impact of restrictions on FDI on technology transfer is much smaller than the impact of the strictness of patent law enforcement on technology transfer.

¹⁷ Because the variable is logged, the coefficient can be interpreted as an elasticity (see [Cameron and Trivedi 2010](#)).

¹⁸ The coefficient of continuous variables in Poisson estimations can be interpreted as a semi-elasticity (again, see [Cameron and Trivedi 2010](#)). Semi-elasticities as reported by the margins command in Stata indeed give extremely close results.

Table 5 Comparison with non-climate technologies

Technology	(1) Climate Tech	(2) All Tech
$\ln k_{jt-1}$	-0.173 (0.108)	-0.113 (0.128)
ipr_{jt}	0.543*** (0.056)	0.513*** (0.052)
$fdi_control_{jt}$	-0.064*** (0.013)	-0.065*** (0.013)
$tariff_{jt}$	-0.095** (0.041)	-0.089** (0.038)
$policy_{jt}$	0.002 (0.002)	0.001 (0.001)
I_{it-1}	0.584*** (0.067)	0.220** (0.096)
$\ln GDP_{jt}$	1.031** (0.444)	0.872** (0.409)
Observations	26,950	51,975
Country pairs	2,450	4,725
R2	0.94	0.95

* Significant at the 10 % level, ** significant at the 5 % level, *** significant at the 1 % level. The dependent variable is the number of patents transferred from country i to country j in year t. All columns are estimated using a fixed-effects Poisson and include a full set of year dummies (not reported for the sake of brevity). Standard errors clustered at country-pair level are in parentheses

6.5 Do climate policies promote technology transfer?

We find evidence of a positive answer in five regressions, in line with expectations. A possible explanation for the lack of significant coefficients in the other regressions could be that our variable—the count of climate policies in the recipient country—is an imperfect proxy of the strictness of climate policies in the different technology areas.

6.6 Do climate technologies differ from other technologies?

We also investigate whether the international transfer of climate technologies behave differently from the international transfer other types of technologies. If climate technologies are specific, this could justify specific adjustments of policies governing trade, IPR or FDI. In Table 5 we compare two models: In column 1, we use the number of all climate-related patents as the dependent variable; in column 2, we use the number of patent transfers in all technologies as the dependent variable. This dependent variable is constructed based on the 60 million patents available from PATSTAT. Although, arguably, the variable $policy_{jt}$ does not influence the international flow of non-climate technologies, we keep it in the second model to avoid any bias in the coefficients of ipr_{jt} , $tariff_{jt}$, and $fdi_control_{jt}$ which could arise from spurious correlations with $policy_{jt}$. We find that the coefficients of the policy variables are very similar across the two models. Thus, we conclude that climate-friendly technologies do not differ from other technologies.

7 Conclusion

In this paper we use the PATSTAT database to analyze the international diffusion of patented inventions in 10 climate-related technologies between 1995 and 2007. We draw conclusions about the factors that promote or hinder international technology transfer.

Regression results show that technological capacity of recipient countries actually reduces the transfer of foreign technologies. Although surprising at first glance, the result actually reflects the fact that these technological capacities function as both complements to and substitutes for foreign inventions. We proxy the technological capacity with the stock of patents previously filed in the recipient country by local inventors (as is usually done in the literature). Technological capacity then functions as a complement to foreign invention because local technology users are better equipped to absorb foreign technologies. But technological capacity can also *reduce* the transfer of foreign technologies because local technologies can function as a substitute for foreign technologies.

We also assess the impacts of different policy barriers. The results stress that restrictions to the international trade of equipment goods (high tariff rates) and barriers to FDI both negatively impact the international diffusion of patented knowledge. Lax intellectual property laws have the same negative impact for most technologies. However, the size of the impact differs. Depending on the technology, on a common 1-to-10 scale, a 1-unit increase in trade leads to 7–15 % fewer imports, while a 1-unit increase in FDI barriers leads to 4.5–8 % fewer imports. By contrast, reducing IPR strictness by one unit cuts transfers by 27–60 %.

Finally, as expected, climate policies have a positive impact, but the positive impact is significant in only five technologies. This result probably stems from the quality of the variable, that is, because we only counted the number of policies in place, we did not accurately evaluate the strictness of these policies.

Our results have clear policy implications. First, relaxing IPR for green technologies, as advocated by certain developing countries, could be detrimental to the international diffusion of technological knowledge. This claim is reinforced by the fact that, if IPR is weakened, it prompts innovators to rely on secrecy to protect their inventions, which would negatively impact the international diffusion of knowledge because secret inventions diffuse less extensively in the recipient economy. Similarly, raising barriers to trade or to FDI also could be detrimental to international technology diffusion, although the impact may not be as strong.

Another policy message contained in our results is that climate technologies do not respond differently than the average technology to changes in IPR or trade or FDI policies. This suggests that there is little ground to design specific policies on the supply side in this area. If any, the specificity of climate technologies rather lies on the demand side. Because they represent cleaner alternatives to established technologies, their profitability largely depends on the existence of environment policies that internalize the social cost of pollution.

Finally, the fact that stronger local technological capacities tend to decrease technology flows should not lead to the disregard for capacity building policies. Indeed, we interpret this result as suggesting that higher technological absorptive capabilities also mean more local innovation, which can substitute for technology imports.

Appendix: List of countries included in the dataset

Algeria, Armenia, Argentina, Austria, Australia, Azerbaijan, Bosnia and Herzegovina, Belgium, Bulgaria, Brazil, Belarus, Canada, Croatia, Chile, China, Colombia, Costa Rica,

Czechoslovakia, Cuba, Cyprus, Czech Republic, Germany, Denmark, Dominican Republic, Ecuador, Estonia, Egypt, Spain, Finland, France, Georgia, Ghana, Greece, Guatemala, Hungary, Indonesia, Ireland, Israel, India, Iceland, Italy, Japan, Kenya, Kyrgyzstan, Korea, Kazakhstan, Sri Lanka, Lithuania, Luxembourg, Latvia, Morocco, Monaco, Moldova, Macedonia, Mongolia, Malta, Malawi, Mexico, Malaysia, Nicaragua, Netherlands, Norway, New Zealand, Panama, Peru, Philippines, Pakistan, Poland, Portugal, Paraguay, Romania, Russia, Sudan, Sweden, Singapore, Slovenia, Slovakia, San Marino, Salvador, Switzerland, Syria, Thailand, Tajikistan, Turkey, Taiwan, Ukraine, United Arab Emirates, United Kingdom, United States, Uruguay, Uzbekistan, Venezuela, Vietnam, South Africa, Zambia, Zimbabwe

References

- Anand B, Khanna T (2000) The structure of licensing contracts. *J Ind Econ* 48((1):103–135
- Arora A, Ceccagnoli M, Cohen WM (2008) R&D and the patent premium. *Int J Ind Org* 26(5):1153–1179
- Barton J (2007) Intellectual property and access to clean energy technologies in developing countries. An analysis of solar photovoltaic, biofuel and wind technologies. ICTSD Programme on Trade and Environment, Issue Paper No. 2
- Branstetter L, Fisman R, Foley F (2006) Do stronger intellectual property rights increase international technology transfer? Empirical evidence from U.S. firm-level panel data. *Q J Econ* 121(1):321–349
- Cameron AC, Trivedi PK (2010) Microeconometrics using stata, revised edition. Stata Press books, StataCorp LP
- Coe D, Helpman E (1995) International R&D spillovers. *Eur Econ Rev* 39(5):859–887
- Cohen WM, Nelson RR, Walsh JP (2000) Protecting their Intellectual Assets: Appropriability Conditions and Why U.S. Manufacturing Firms Patent (or Not). NBER Working Paper 7552
- Dechezleprêtre A, Glachant M, Hascic I, Johnstone N, Ménière Y (2011) Invention and transfer of climate change mitigation technologies: a global analysis. *Rev Env Econ Policy* 5(1):109–130
- Dela Tour A, Glachant M, Ménière Y (2011) Innovation and international technology transfer: the case of the chinese photovoltaic industry. *Energy Policy* 39(2):761–770
- Dekker T, Vollebergh HRJ, De Vries FP, Withagen C (2012) Inciting protocols. *J Environ Econ Manage* 64(1):45–67
- Dernis H, Guellec D (2001) Using patent counts for cross-country comparisons of technology output. *STI Rev* 27:129–146
- Eaton J, Kortum S (1996) Trade in ideas patenting and productivity in the OECD. *J Int Econ* 40(3–4):251–278
- Eaton J, Kortum S (1999) International technology diffusion: theory and measurement. *Int Econ Rev* 40(3):537–570
- Frietsch R, Schmoch U (2006) Technological structures and performance as reflected by patent indicators. In: Schmoch U, Rammer C, Legler H (eds) National systems of innovation in comparison. Structure and performance indicators for knowledge societies. Springer, Dordrecht
- Griliches Z (1990) Patent statistics as economic indicators: a survey. *J Econ Lit* 28(4):1661–1707
- Hascic I, Johnstone N (2011) CDM and international technology transfer: empirical evidence on wind power. *Clim Policy* 11(6):1303–1314
- Helfgott S (1993) Patent filing costs around the world. *J Pat Trademark Off Soc* 75:567–580
- ICSTD (2008) Climate change, technology transfer and intellectual property rights. Background paper of the Trade and Climate Change Seminar, 18–20 June, Copenhagen, Denmark
- Ivus O (2010) Do stronger patent rights raise high-tech exports to the developing world?. *J Int Econ* 81:38–47
- Jaffe A, Trajtenberg M, Henderson R (1993) Geographic localization of knowledge spillovers as evidenced by patent citations. *Q J Econ* 108(3):577–598
- Johnstone N, Haščić I, Popp D (2010) Renewable energy policies and technological innovation: evidence based on patent counts. *Environ Resour Econ* 45(1):133–155
- Keller W (2002) Geographic localization of international technology diffusion. *Am Econ Rev* 92:120–142
- Keller W (2004) International technology diffusion. *J Econ Lit* 42(3):752–782
- Lanjouw JO, Mody A (1996) Innovation and the international diffusion of environmentally responsive technology. *Res Policy* 25:549–571
- Lanjouw J, Pakes A, Putnam J (1998) How to count patents and value intellectual property: uses of patent renewal and application data. *J Ind Econ* 46(4):405–432

- Lichtenberg F, Pottelsberghe Potterie Bvan de la (2000) Does foreign direct investment transfer technology across borders?. *Rev Econ Stat* 83(3):490–497
- Maskus KE (2000) Intellectual property rights in the global economy. Institute for International Economics, Washington, DC
- Ockwell D, Watson J, MacKerron G, Pal P, Yamin F (2008) Key policy considerations for facilitating low carbon technology transfer to developing countries. *Energy Policy* 36(11):4104–4115
- Park WG, Lippoldt DC (2008) Technology transfer and the economic implications of the strengthening of intellectual property rights in developing countries. OECD Trade Policy Working Papers 62, OECD Trade Directorate
- Peri G (2005) Determinants of knowledge flows and their effect on innovation. *Rev Econ Stat* 87(2):308–322
- Santos Silva JMC, Tenreiro S (2011) Further simulation evidence on the performance of the Poisson pseudo-maximum likelihood estimator. *Econ Lett* 112(2):220–222
- Smith PJ (2001) How do foreign patent rights affect U.S. exports, affiliate sales, and licenses?. *J Int Econ* 55:411–439
- Verdolini E, Galeotti M (2011) At home and abroad: an empirical analysis of innovation and diffusion in energy technologies. *J Environ Econ Manag* 61(2):119–134